

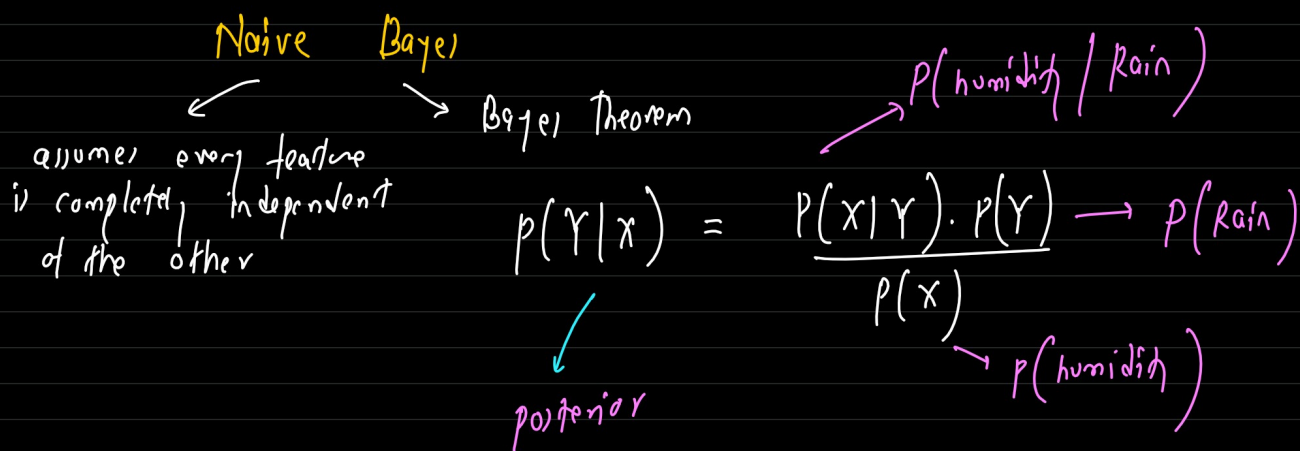
05-07-2026

Agenda: ML-15

- Naive Bayes Algo (2 methods)

Naive Bayes Algo:

A probabilistic machine learning algo. It starts with a baseline assumption/guess & then updates its belief every time it sees a new piece of information (features), multiplying the guess together to reach the final verdict.



$Y \rightarrow \text{weather (Rain)}$

$X \rightarrow \text{humidity}$

$X_1 \rightarrow \text{wind speed}$

$X_2 \rightarrow \text{cloud}$

$p(\text{Rain} | \text{humidity})$

Base assumption

$$\text{Total_score} = p(\text{Rain}) \times p(\text{humidity} | \text{Rain}) \times \boxed{p(\text{wind speed} | \text{Rain})} \times p(\text{cloud} | \text{Rain})$$

why Naive Bayes?

- Data is categorical
- we want probabilistic reasoning
- we want something that learns from frequency count
- we want fast & interpretable solution.

NB → classification → probabilistic model
→ supervised ML Algo

Dataset: (Tennis)

outlook	Temp	play
Sunny	Hot	No
Sunny	Mild	No
Cloudy	Hot	Yes
Rainy	Mild	Yes
Rainy	Cool	Yes
Cloudy	Cool	Yes

prob of Rainy & Hot?

Baseline Assumption:

$$p(\text{Yes}) = \frac{4}{6} = 0.66$$

$$p(\text{No}) = \frac{2}{6} = 0.33$$

	outlook	outlook - calc	Temp	Temp - calculation
$p(\text{Yes}) \rightarrow 0.66$	cloudy	$2/4 \rightarrow 0.5$	Hot	$1/4 \rightarrow 0.25$
	Rainy	$2/4 \rightarrow 0.5$	mild	$1/4 \rightarrow 0.25$
	Sunny	$0/4 \rightarrow 0$	cool	$2/4 \rightarrow 0.5$
$p(\text{No}) \rightarrow 0.33$	cloudy	$0/2 \rightarrow 0$	Hot	$1/2 \rightarrow 0.5$
	Rainy	$0/2 \rightarrow 0$	mild	$1/2 \rightarrow 0.5$
	Sunny	$2/2 \rightarrow 1$	cool	$0/2 \rightarrow 0$

$$\begin{aligned}
 \underline{P(\text{Yes} \mid \text{Rain, Hot})} &\propto P(\text{Yes}) \times P(\text{Rain} \mid \text{Yes}) \times P(\text{Hot} \mid \text{Yes}) \\
 &= 0.66 \times 0.5 \times 0.25 \\
 &= 0.0825
 \end{aligned}$$

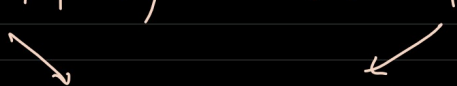
$$\begin{aligned}
 P(\text{No} \mid \text{Rain, Hot}) &\propto P(\text{No}) \times P(\text{Rain} \mid \text{No}) \times P(\text{Hot} \mid \text{No}) \\
 &= 0.33 \times 0 \times 0.5 \\
 &= 0
 \end{aligned}$$

$$P(\text{Yes} \mid \text{Rain, Hot}) > P(\text{No} \mid \text{Rain, Hot}) = \underline{\underline{\text{Yes}}}$$

→ prob(play) | Sunny & cool

$$\begin{aligned}
 P(\text{Yes} \mid \text{Sunny, cool}) &\propto P(\text{Yes}) \times P(\text{Sunny} \mid \text{Yes}) \times P(\text{cool} \mid \text{Yes}) \\
 &= 0.66 \times 0 \times 0.5 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 P(\text{No} \mid \text{Sunny, cool}) &\propto P(\text{No}) \times P(\text{Sunny} \mid \text{No}) \times P(\text{cool} \mid \text{No}) \\
 &= 0.33 \times 1 \times 0 \\
 &= 0
 \end{aligned}$$

$P(\text{Yes} \mid \text{Sunny, cool})$ $P(\text{No} \mid \text{Sunny, cool})$

 Both are 0 → zero frequency prob

To solve above, we use something called a smoothing

$$\frac{\text{numerator} + 1}{\text{denominator} + (\text{no. of categories})}$$

→ we add 1 to every count — numerator

→ divide by # samples + # categories

to keep prob balance

100,000

outlook	outlook_calc	outlook_new	Temp	Temp_calculation	temp_new
cloudy	$\frac{2}{4} \rightarrow 0.5$	$\frac{2+1}{4+3} \rightarrow 0.429$	Hot	$\frac{1}{4} \rightarrow 0.25$	$\frac{1+1}{4+3} \rightarrow 0.286$
Rainy	$\frac{2}{4} \rightarrow 0.5$	$\frac{2+1}{4+3} \rightarrow 0.429$	mild	$\frac{1}{4} \rightarrow 0.25$	$\frac{1+1}{4+3} \rightarrow 0.286$
Sunny	$\frac{0}{4} \rightarrow 0$	$\frac{0+1}{4+3} \rightarrow 0.143$	cool	$\frac{2}{4} \rightarrow 0.5$	$\frac{2+1}{4+3} \rightarrow 0.429$
cloudy	$\frac{0}{2} \rightarrow 0$	$\frac{0+1}{2+3} \rightarrow 0.2$	Hot	$\frac{1}{2} \rightarrow 0.5$	$\frac{1+1}{2+3} \rightarrow 0.4$
Rainy	$\frac{0}{2} \rightarrow 0$	$\frac{0+1}{2+3} \rightarrow 0.2$	mild	$\frac{1}{2} \rightarrow 0.5$	$\frac{1+1}{2+3} \rightarrow 0.4$
Sunny	$\frac{2}{2} \rightarrow 1$	$\frac{2+1}{2+3} \rightarrow 0.6$	cool	$\frac{0}{2} \rightarrow 0$	$\frac{0+1}{2+3} \rightarrow 0.2$

$$p(\text{Yes}) \rightarrow \frac{9}{16} \rightarrow 0.56$$

$$p(\text{No}) \rightarrow \frac{2}{6} \rightarrow 0.33$$

→ $\text{prob}(p/q) \mid \text{Junny} \& \text{cool}$

$$\begin{aligned} p(\text{Yes} \mid \text{Junny}, \text{cool}) &\propto p(\text{Yes}) \times p(\text{Junny} \mid \text{Yes}) \times p(\text{cool} \mid \text{Yes}) \\ &= 0.66 \times 0.143 \times 0.429 \\ &= 0.040 \end{aligned}$$

$$\begin{aligned} p(\text{No} \mid \text{Junny}, \text{cool}) &\propto p(\text{No}) \times p(\text{Junny} \mid \text{No}) \times p(\text{cool} \mid \text{No}) \\ &= 0.33 \times 0.6 \times 0.2 \\ &= 0.0396 \end{aligned}$$

$$p(\text{Yes} \mid \text{Junny}, \text{cool}) \quad p(\text{No} \mid \text{Junny}, \text{cool})$$

Normalise to get the final prob:-

$$\text{sum} = 0.04 + 0.039 = 0.0796$$

$$p(\text{Yes} \mid \text{Junny}, \text{cool}) = \frac{0.04}{0.0796} \approx 0.5025$$

$$p(\text{No} \mid \text{Junny}, \text{cool}) = \frac{0.039}{0.0796} \approx 0.4974$$

$$p(\text{Yes} \mid \text{Junny}, \text{cool}) > p(\text{No} \mid \text{Junny}, \text{cool})$$

feature1 → categorical
target → categorical → categorical Naive Bayes

Next topic:-

Dataset $\rightarrow$ numerical features $\rightarrow$ categorical NB?
 $\rightarrow$ Categorical Target column $\rightarrow$

height (cm) Taller / Not
 Yes
 No

$\rightarrow$ Gaussian Naive Bayes
 $\downarrow$
Assume
the continuous
numbers follow a
Gaussian (normal) distribution

$$\text{prob} = \frac{1}{\infty}$$

step 1: learn the distribution for each class

$\swarrow$ $\searrow$
mean (μ) variance (σ^2)

step 2: Gaussian PDF formula (finding the likelihood)

$$\underset{\substack{\text{likelihood} \\ \text{of } x}}{P(x|Y)} = \underset{\substack{\text{given } Y}}{=} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = N(x, \mu, \sigma^2)$$

$\swarrow \quad \searrow$
 $\frac{22}{7} \mid 3.14$ $e \rightarrow 2.718$

we compute:

$$P(Y|x) \propto P(Y) \prod N(x, \mu, \sigma^2)$$

$\swarrow$
product
operator

$\swarrow$
for loop for
multiplication

Data set:

step 1: mean, variance, std

id	x_1	x_2	y
A	1.0	2.0	yes
B	2.0	1.0	yes
C	1.5	1.8	yes
D	3.0	3.0	no
E	3.5	2.8	no
F	2.9	3.2	no

$$\mu(x_1 = \text{yes}) = 1 + 2 + 1.5 / 3 \rightarrow 1.5 \leftarrow \mu_{\text{yes}, 1}$$

$$\mu(x_1 = \text{no}) = 3 + 3.5 + 2.9 / 3 \rightarrow 3.133 \rightarrow \mu_{\text{no}, 1}$$

$$\mu(x_2 = \text{yes}) = 2 + 1 + 1.8 / 3 \rightarrow 1.6 \rightarrow \mu_{\text{yes}, 2}$$

$$\mu(x_2 = \text{no}) = 3 + 2.8 + 3.2 / 3 \rightarrow 3.0 \rightarrow \mu_{\text{no}, 2}$$

$$\sigma^2 = \sum (x - \mu)^2 / n \leftarrow \text{variance}$$

$$\begin{aligned} (\text{yes})(x_1) &= (1 - 1.5)^2, (2 - 1.5)^2, (1.5 - 1.5)^2 \\ &= (0.5)^2, (0.5)^2, (0)^2 \end{aligned}$$

$$\text{sum} = 0.25 + 0.25 + 0 \rightarrow 0.5 / 3 \rightarrow 0.166$$

$$(\text{no})(x_1) = \text{sum} \rightarrow 0.2066 \rightarrow 0.2066 / 3 \rightarrow 0.0688$$

$$(\text{yes})(x_2) = \text{sum} \rightarrow 0.56 \rightarrow 0.56 / 3 \rightarrow 0.186$$

$$(\text{no})(x_2) = \text{sum} \rightarrow 0.08 \rightarrow 0.08 / 3 \rightarrow 0.0266$$

Step 2

input $\rightarrow x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2.0 \\ 2.0 \end{pmatrix}$

$\rightarrow$ calculate gaussian pdf for each feature

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$\begin{aligned} (\text{yes})(x_1 = 2.0) &= \frac{1}{\sqrt{2\pi \cdot 0.166}} \cdot \exp\left(-\frac{(2 - 1.5)^2}{2 \cdot 0.166}\right) \\ &\quad \swarrow \quad \quad \quad \searrow \\ &\quad \approx 0.9772 \quad \quad \quad 0.472 \end{aligned}$$

$$p(x_1 | y_{e1}) \approx 0.9722 \times 0.472 \approx 0.4616$$

$$p(x_2 | y_{e1}) \approx 0.6009$$

$$p(x_1 | N_0) \approx 1.36 \times 10^{-4}$$

$$p(x_2 | N_0) \approx 1.68 \times 10^{-8}$$

Joint prob:

$$\begin{array}{cc} x_1(y) \cdot x_2(y) & x_1(N) \cdot x_2(N) \\ \swarrow y_{e1} & \searrow N_0 \end{array}$$

$$p(x_1 | N_0) \times p(x_2 | N_0) \rightarrow p(x | N_0)$$

$$1.36 \times 10^{-4} \times 1.68 \times 10^{-8}$$

$$\approx 2.29 \times 10^{-12}$$

$$p(x_1 | y_{e1}) \times p(x_2 | y_{e1}) \rightarrow p(x | y_{e1})$$

$$0.4616 \times 0.6009$$

$$\approx 0.2774$$

step 3:

$$p(y_{e1} | x) = p(y) \cdot p(x | y_{e1}) = 0.5 \times 0.2774 \approx 0.1387$$

$$p(N_0 | x) = p(N) \cdot p(x | N_0) = 0.5 \times 2.29 \times 10^{-12} \approx 1.145 \times 10^{-12}$$

step 4: normalise to get class prob

$$\text{sum} = 0.1387 + 1.145 \times 10^{-12}$$

$$= 0.1387...$$

$$p(\text{Yes} | x) \approx \frac{0.1387}{0.1387...} \approx 0.99999 \dots$$

$$p(\text{No} | x) \approx \frac{1.145 \times 10^{-12}}{0.1387...} \approx 8.26 \times 10^{-12}$$

Decision : predict $\rightarrow \text{Yes}$ (very strongly)

10 features $\rightarrow x_1, \dots, x_{10}$

$$p(\text{Yes} | x) = \underbrace{p(\text{Yes})}_{0.025} \times \underbrace{p(x_1 | \text{Yes})}_{0.3} \times \underbrace{p(x_2 | \text{Yes})}_{0.9} \times \dots \times \underbrace{p(x_{10} | \text{Yes})}_{0.1}$$

small decimal number

0.0000000000000000...

computers have a limit on how small a decimal they can store.

$$p(\text{Yes} | x) = \log \left(p(\text{Yes}) \times p(x_1 | \text{Yes}) \times p(x_2 | \text{Yes}) \times \dots \times p(x_{10} | \text{Yes}) \right)$$

$$\log\text{-score} = \log \left(p(\text{Yes}) \times p(x_1 | \text{Yes}) \times p(x_2 | \text{Yes}) \times \dots \times p(x_{10} | \text{Yes}) \right)$$

$$\log(A \times B) = \log(A) + \log(B)$$

$$\log\text{-score} = \log \left(p(\text{Yes}) \right) + \log \left(p(x_1 | \text{Yes}) \right) + \log \left(p(x_2 | \text{Yes}) \right) + \dots + \log \left(p(x_{10} | \text{Yes}) \right)$$

numerical stability

problem statement:

features $\rightarrow$ category $\rightarrow$ categorical Naive Bayes
 $\rightarrow$ numerical $\rightarrow$ Gaussian Naive Bayes

x_1 (categorical)
(weather)

x_2 (numerical)
(temperature)

y (Target)
(play)

$$p(y | \text{sunny}, 72^\circ\text{F})$$

$$p(y_{\text{es}} | \text{sunny}, 72^\circ\text{F}) \propto p(y_{\text{es}}) \times p(\text{sunny} | y_{\text{es}}) \times p(72^\circ\text{F} | y_{\text{es}})$$

$$p(\text{No} | \text{sunny}, 75^\circ\text{F}) \propto p(\text{No}) \times p(\text{sunny} | \text{No}) \times p(75^\circ\text{F} | \text{No})$$

$\swarrow$ categorical likelihood $\swarrow$ numerical likelihood

Assignment:

sklearn $\rightarrow$ categorical NB
 $\rightarrow$ Gaussian NB

Assignment: Think, how to solve with Naive Bayes when features are both Categorical & Numerical using either Categorical/Gaussian NB

mixed-naive-bayes $\rightarrow$ Mixed NB